ML FOR LANGUAGE PROCESSING | EM40001

IMAGE-TO-TEXT DESCRIPTION GENERATION

Using pretrained BLIP Base and BLIP Large models

Group 16
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1. Problem Statement & Objectives

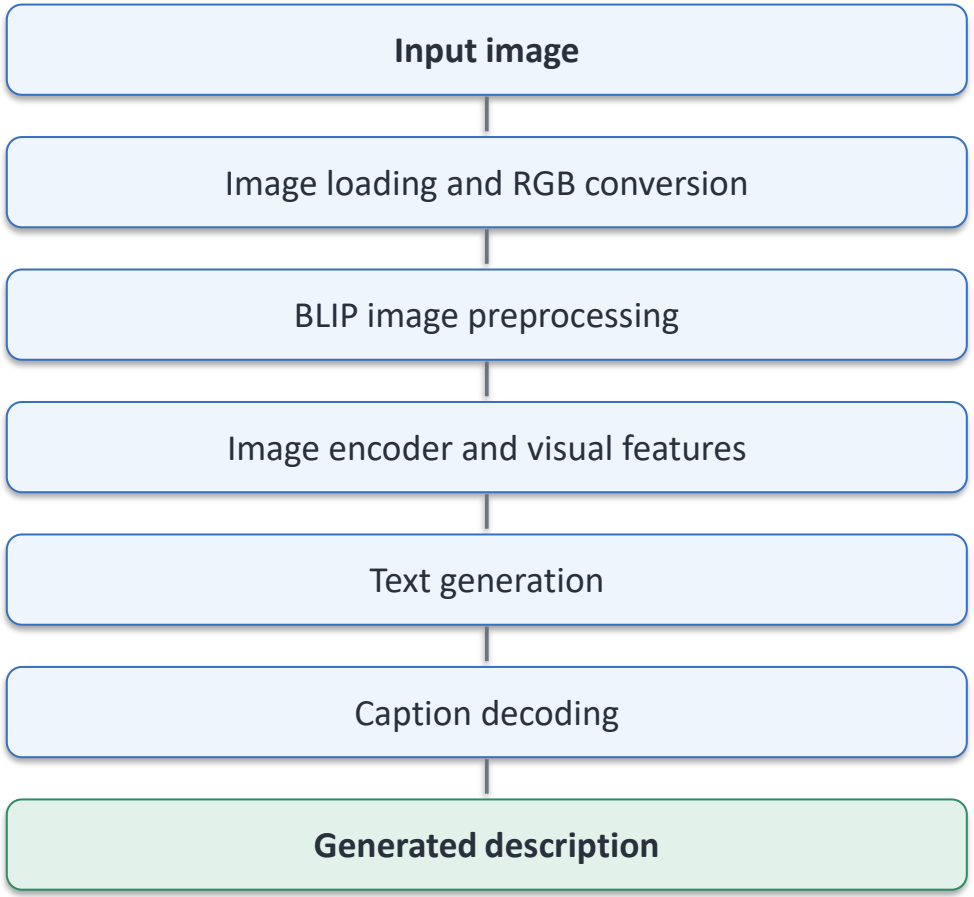
Problem Statement

Understanding an image requires identifying its visual content and expressing it in natural language. Manual image description can require time and effort.

Objectives

- Generate descriptions using pretrained BLIP Base and BLIP Large
- Use the MS COCO 2017 Validation Dataset
- Evaluate captions with BLEU, METEOR and ROUGE-L
- Compare model performance and limitations

2. Proposed Processing Pipeline



3. Models & Implementation

BLIP Base

Salesforce/blip-image-captioning-base
Smaller model configuration.

BLIP Large

Salesforce/blip-image-captioning-large
Larger model configuration.

- Environment: Google Colab (GPU or CPU; NVIDIA T4 when available)
- Libraries: Python, PyTorch, Hugging Face Transformers, Pillow, NumPy, NLTK, ROUGE Score and Matplotlib
- Images are loaded, converted to RGB and prepared by the corresponding BLIP processor.
- Both models use the same caption-generation settings and evaluation dataset.

4. Demonstration: Sample Inputs & Outputs



Fig 5.1. Sample Image 1 and Generated Descriptions by BLIP Base and BLIP Large.



Fig 5.2. Sample Image 1 and Generated Descriptions by BLIP Base and BLIP Large.



Fig 5.3. Sample Image 1 and Generated Descriptions by BLIP Base and BLIP Large.



Fig 5.4. Sample Image 1 and Generated Descriptions by BLIP Base and BLIP Large.

5. Results & Model Comparison

Evaluation / Runtime	BLIP Base	BLIP Large
BLEU-1	0.6327	0.6763
BLEU-2	0.5031	0.5202
BLEU-4	0.3053	0.2898
METEOR	0.4239	0.5284
ROUGE-L	0.5381	0.5394
Average inference time	0.2578 s	0.4704 s
Throughput	3.88 img/s	2.13 img/s
Model parameters	223,971,600	446,290,500
Peak GPU memory	1,096.56 MB	1,952.90 MB

6. Usability & Limitations

Practical Applications

- Accessibility support with text-to-speech
- Image search and retrieval
- Digital content organisation
- Educational materials
- Social media and web content
- Automated image documentation

System Limitations

- May misunderstand complex scenes or relationships
- Captions may omit details or contain errors
- Usually generates short descriptions
- Evaluation depends on the dataset and reference captions
- BLIP Large requires more computation

7. Group Contributions

Ayushman Mahapatra | Roll No. 2330020

Contribution: Contribution: Dataset Collection and Preparation, Image Preprocessing, and Input Image Management.

Shourya Basu | Roll No. 2330400

Contribution: BLIP Model Integration, including implementation of BLIP Base and BLIP Large for image caption generation.

Shrutik Sinha | Roll No. 2330404

Contribution: Model Evaluation, Performance Analysis, and Comparison of Generated Captions using BLEU, METEOR, and ROUGE-L metrics.

Manit Kumar Sahu | Roll No. 2430605

Contribution: Project Documentation, Results Compilation, Presentation Preparation, and Demonstration.

Project Links

GitHub Repository

https://github.com/Shourya-Basu/Image-to-Text_Description_Generation

Demonstration Video

https://www.youtube.com/watch?v=0f6SnVhzlEM

MS COCO 2017 Dataset

https://cocodataset.org/#download

BLIP Base Model

Salesforce/blip-image-captioning-base

BLIP Large Model

Salesforce/blip-image-captioning-large

Project Notebook

https://colab.research.google.com/drive/1iX2zN8-TGlNTQB_lCS52HfXuTYOl5LBE?usp=sharing

Thank you